XRPL Validator — Daily Engineering Report · 2026-11-04

Period: 2026-11-04 00:00 to 24:00 US Eastern (2026-11-04 05:00 to 2026-11-05 05:00 UTC); ledgers #108,103,116 to #108,124,748. Every count from the ledger and the validator's logs stops at #108,124,748, closed 04:59:52 UTC. Generated 2026-11-05 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — coverage outside planned maintenance 99.009% (objective 99.9%): 214 ledgers unsigned; 3 engine receipts open.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,380 of the 21,633 ledgers the network closed (98.830%); every validation matched the network's validated ledger. Unsigned: 39 during planned restarts, 214 outside maintenance (7 incidents, 2 isolated misses).
- **Native engine:** 9 engine receipts: #108,119,206 — fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); #108,119,208 — fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); #108,119,211 — fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); #108,120,169 — fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); #108,120,171 — fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); #108,120,173 — fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); #108,120,293 — open; #108,120,297 — open; #108,120,300 — triaged: an isolated replay reproduced the difference.
- **Changes:** 1 deployment (cycle 191), 5 commits merged into main.
- **Network:** 3 slow-close periods; the longest, 07:58 UTC–07:58 UTC, closed ledgers up to 18 s apart.
- **Independence program:** 7 of 30 qualifying days (snapshot at 05:20:44 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,633

Measure	Value
Validations issued	21,380
Coverage (validations / ledgers closed)	98.830%
Validations matching the network's validated ledger hash	21,380 of 21,380
Unsigned during planned restarts	39
Unsigned outside maintenance	214
Coverage outside planned maintenance (objective 99.9%)	99.009%
Restarts	1
Basis: our own ledger hash	21,380
External agreement, data.xrpl.org (UTC day 2026-11-04)	0.99812 (39 missed of 20,732)

Restarts:

- 00:49:43 UTC: deployment of cycle 191; validations resumed at #108,121,002 (39 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,103,116–#108,124,748), by step:

Step	Ledgers with a receipt
Step 1 · native engine	9
Step 2 · records	9
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	9
Writer	9

Live counters, snapshot at 05:20:44 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	4,040 ledgers, 814,208 transactions	4,040 fully matched; results 0, bytes 0, keys 0/0
2 · transaction records	4,051 ledgers, 808,861 records	808,861 exact; differ 0, tree 0
3 · transaction order	4,094 ledgers (mode: on)	4,094 exact; order 0, results 0, bytes 0, refused 0
4 · ledger data from peers	4,100 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	4,095 ledgers judged	4,095 agreed; differ 0, short of quorum 0
5 · our own ledger hash	4,098 ledgers	4,051 own; differ 0, partial 47
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	22:51 UTC	#108,119,206	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx 32631E3621BE: RippleState 75B30408 (modified on the network): absent from our record	fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); see INC-01
R-02	22:51 UTC	#108,119,208	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx 2D94D55DE2A4: RippleState DDB66A1D (modified on the network): absent from our record	fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); see INC-01
R-03	22:51 UTC	#108,119,211	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx 97AFE67D5E53: RippleState B33E52DD (modified on the network): FinalFields.HighQualityIn ours=null theirs=1000000000	fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); see INC-02
R-04	23:55 UTC	#108,120,169	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx 5AAD06D2B52B: RippleState 75B30408 (modified on the network): extra	fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); see INC-03
R-05	23:55 UTC	#108,120,171	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx 72C77618E1AF: RippleState DDB66A1D (modified on the network): extra	fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); see INC-03
R-06	23:55 UTC	#108,120,173	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx B4B08288D89F: RippleState B33E52DD (modified on the network): FinalFields.HighQualityIn ours=null theirs=1000000000	fixed: F422 TrustSet stores a QualityIn of QUALITY_ONE, only QualityOut folds to 0 (b33685d), live since 11-04 (cycle 191); see INC-03
R-07	00:03 UTC	#108,120,293	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx BDE1CD614BC9: RippleState 75B30408 (modified on the network): extra	open
R-08	00:04 UTC	#108,120,297	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx 4F0021FA01F1: RippleState DDB66A1D (modified on the network): extra	open
R-09	00:04 UTC	#108,120,300	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	TrustSet tx D88546693F83: RippleState B33E52DD (modified on the network): FinalFields.HighQualityIn ours=null theirs=1000000000	triaged: an isolated replay reproduced the difference; no fix recorded for this ledger; see INC-05

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-11-04-01, INC-01 is INC-2026-11-04-01.

Incidents

INC-01 · 4 ledgers unsigned

Window	22:51:12 UTC to 22:51:30 UTC
Ledgers	#108,119,206–#108,119,209 (4)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued

Related receipts	R-01, R-02
Cause	TrustSet.cpp:337-341 folds QUALITY_ONE to 0 for QualityOut only; uQualityIn keeps it, so the existing-line set path and trustCreate store a QualityIn of 1e9 on the sender's side, and a zero-limit new line that sets it is not tecNO_LINE_REDUNDANT. We treated both fields alike and cleared it.
Action	commit b33685d — fix(trust): a QualityIn of QUALITY_ONE is stored, only QualityOut folds to 0; deployed in cycle 191 at 2026-11-05 00:50 UTC

INC-02 · 35 ledgers unsigned

Window	22:51:32 UTC to 22:53:50 UTC
Ledgers	#108,119,211–#108,119,245 (35)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-03
Cause	Cause not recorded.

INC-03 · 40 ledgers unsigned

Window	23:55:30 UTC to 23:58:00 UTC
Ledgers	#108,120,169–#108,120,208 (40)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-04, R-05, R-06
Cause	Cause not recorded.

INC-04 · 8 ledgers unsigned

Window	23:58:12 UTC to 23:58:41 UTC
Ledgers	#108,120,212–#108,120,219 (8)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

INC-05 · 109 ledgers unsigned

Window	00:04:20 UTC to 00:11:31 UTC
Ledgers	#108,120,300–#108,120,408 (109)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-09
Cause	Cause not recorded.

INC-06 · 6 ledgers unsigned

Window	00:13:20 UTC to 00:13:40 UTC
Ledgers	#108,120,435–#108,120,440 (6)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

INC-07 · 8 ledgers unsigned

Window	00:54:01 UTC to 00:54:30 UTC
Ledgers	#108,121,028–#108,121,035 (8)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

Isolated unsigned ledgers (shorter than 3 ledgers, cause not recorded): #108,120,293–#108,120,294 (2) at 00:03:50 UTC, #108,120,297–#108,120,298 (2) at 00:04:10 UTC.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
191	`b33685d`	00:50:33 UTC	2,216 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	399 matched, 0 mismatched	#66 from 01:17 UTC: in progress

Cycle 191 delivered: `b57328d` fix(ledger): deleting a legacy SignerList gives back 2 + its entries; `3b4b64c` feat(writer): a hash failure patches the mirror in place; the breaker is an alarm only; `cccdec7` fix(amm): AMMCreate checks auth, freeze and DefaultRipple before funds; `c7d3f67` fix(the validator's ledger feed): the FFI leg skips pseudo-txs from peers too; `b33685d` fix(trust): a QualityIn of QUALITY_ONE is stored, only QualityOut folds to 0

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
23:58 UTC	xrpl-validator	`b33685d`	fix(trust): a QualityIn of QUALITY_ONE is stored, only QualityOut folds to 0
23:58 UTC	xrpl-validator	`c7d3f67`	fix(the validator's ledger feed): the FFI leg skips pseudo-txs from peers too
23:58 UTC	xrpl-validator	`cccdec7`	fix(amm): AMMCreate checks auth, freeze and DefaultRipple before funds
23:58 UTC	xrpl-validator	`3b4b64c`	feat(writer): a hash failure patches the mirror in place; the breaker is an alarm only
23:58 UTC	xrpl-validator	`b57328d`	fix(ledger): deleting a legacy SignerList gives back 2 + its entries

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(trust): a QualityIn of QUALITY_ONE is stored, only QualityOut folds to 0** (`b33685d`, xrpl-validator) TrustSet.cpp:337-341 folds QUALITY_ONE to 0 for QualityOut only; uQualityIn keeps it, so the existing-line set

path and trustCreate store a QualityIn of 1e9 on the sender's side, and a zero-limit new line that sets it is not tecNO_LINE_REDUNDANT. We treated both fields alike and cleared it.

- **fix(the validator's ledger feed): the FFI leg skips pseudo-txs from peers too** (`c7d3f67`, xrpl-validator) Two FFI-leg divergences on the live validator (cycle 190), the first UNLModify ever: #108012288 (disable xrp.unic.ac.cy) and #108020224 (re-enable jon-nilsen.no), both tefFAILURE where the network had tesSUCCESS. Signing was untouched: the native engine rotates the NegativeUNL at each flag ledger and our own hash matched both.
- **fix(amm): AMMCreate checks auth, freeze and DefaultRipple before funds** (`cccdec7`, xrpl-validator) #108008726 34C1C80B45F6: a PLX/BEEP AMMCreate by an account whose BEEP line its issuer froze (lsfLowFreeze, the issuer being the low side). rippled's preclaim runs requireAuth on both amounts, then checkFrozen on both, then terNO_RIPPLE for an issuer without lsfDefaultRipple, all before the reserve and funds (AMMCreate.cpp:107-146). We had none of the three, so the frozen side counted as zero funds and we answered …
- **feat(writer): a hash failure patches the mirror in place; the breaker is an alarm only** (`3b4b64c`, xrpl-validator) #107989363 (10-29): our bytes failed the state hash, the writer retried from the network's bytes, and the mirror was dropped and rebuilt from state.rocks: 246 s, and with 5a's breaker off (cycle 190) every ledger in them went unsigned, 40 in all.
- **fix(ledger): deleting a legacy SignerList gives back 2 + its entries** (`b57328d`, xrpl-validator) #107989363 0A9A6F552C51 (SignerListSet, SignerQuorum 0) deleted a 2-entry list made before MultiSignReserve: no lsfOneOwnerCount, so rippled's removeSignersFromLedger gives back 2 + entries. Mainnet's OwnerCount went 14 -> 10; ours went 14 -> 13, because the delete branch always gave back one. The replacement path and AccountDelete already had the legacy rule; the delete branch now uses the same one, through …

Independence program

Qualifying days: **7 of 30** (snapshot at 05:20:44 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-04 — tracked #108,098,625–#108,120,168	Value
Validations	21,505 (21,505 own, 0 borrowed)
Coverage of the tracked ledgers	99.82%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,633 (#108,103,116–#108,124,748)

Measure	Value
Mean close interval	3.99 s
Longest close interval	18 s (before #108,112,735)
Closes at least 15 s apart	3
Empty ledgers	0

- Slow closes 07:58:02 UTC–07:58:20 UTC: ledgers #108,105,842, 1 close at least 15 s apart, up to 18 s.
- Slow closes 07:58:24 UTC–07:58:40 UTC: ledgers #108,105,847, 1 close at least 15 s apart, up to 16 s.
- Slow closes 15:33:22 UTC–15:33:40 UTC: ledgers #108,112,735, 1 close at least 15 s apart, up to 18 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- fixCleanup3_4_0 was enabled — ledger 108,112,129, 2026-11-04 14:52 UTC

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

- Receipt R-07 (#108,120,293): open, under investigation.
- Receipt R-08 (#108,120,297): open, under investigation.
- Receipt R-09 (#108,120,300): open, under investigation.
- Unsigned #108,120,293–#108,120,294 (2) (00:03:50 UTC): cause not recorded.
- Unsigned #108,120,297–#108,120,298 (2) (00:04:10 UTC): cause not recorded.
- INC-02 (#108,119,211–#108,119,245 (35)): cause not recorded.
- INC-03 (#108,120,169–#108,120,208 (40)): cause not recorded.
- INC-04 (#108,120,212–#108,120,219 (8)): cause not recorded.
- INC-05 (#108,120,300–#108,120,408 (109)): cause not recorded.
- INC-06 (#108,120,435–#108,120,440 (6)): cause not recorded.
- INC-07 (#108,121,028–#108,121,035 (8)): cause not recorded.
- Soak #66 (cycle 191) in progress since 2026-11-05 01:17 UTC; due to complete 2026-11-06 01:17 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
11-04 23:06	11-04 23:58	xrpl-validator	`b33685d`	fix(trust): a QualityIn of QUALITY_ONE is stored, only QualityOut folds to 0
10-31 15:03	11-04 23:58	xrpl-validator	`c7d3f67`	fix(the validator's ledger feed): the FFI leg skips pseudo-txs from peers too
10-30 21:09	11-04 23:58	xrpl-validator	`cccdec7`	fix(amm): AMMCreate checks auth, freeze and DefaultRipple before funds
10-30 00:23	11-04 23:58	xrpl-validator	`3b4b64c`	feat(writer): a hash failure patches the mirror in place; the breaker is an alarm only
10-30 00:10	11-04 23:58	xrpl-validator	`b57328d`	fix(ledger): deleting a legacy SignerList gives back 2 + its entries

Times in UTC. A commit appears on the day it was committed to main or merged into it.